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QUICK REFERENCE · ASQ CERTIFIED RELIABILITY ENGINEER

CRE Formula Encyclopedia

Every formula, organised by Body of Knowledge section

125 reliability formulas: each with the equation, when to use it and the exam trap to avoid. Mapped to the CRE Body of Knowledge and the ASQ CRE Handbook chapter, and colour-coded so you can find the right formula in seconds.

125	63	62	5
FORMULAS	HANDBOOK EQUATIONS	NAMED DISPLAYS	BOK SECTIONS

19	—	56	26	24
I. Fundamentals formulas	**II. Risk Management** Qualitative	**III. Probability & Statistics** formulas	**IV. Planning, Testing & Modeling** formulas	**V. Lifecycle Reliability** formulas

Contents

125 formulas across the five sections of the CRE Body of Knowledge. Each section starts on a new page; the coloured tab on the page edge shows where you are.

I. Fundamentals 19 formulas	Life, failure, repair, and availability from Chapters 1 and 2, including the OEE and Duane displays.	17 numbered · 2 named	**p. 4**
II. Risk Management No formulas	Chapters 3, 4, and 5 describe hazard analysis, FMEA, fault trees, and control selection. This edition does not assign equation numbers in those chapters.	Qualitative	—
III. Probability & Statistics 56 formulas	Probability, life distributions, sampling, capability, and control charts from Chapters 6 and 7.	17 numbered · 39 named	**p. 7**
IV. Planning, Testing & Modeling 26 formulas	Growth, accelerated tests, and reliability models from Chapters 8, 9, and 10.	15 numbered · 11 named	**p. 18**
V. Lifecycle Reliability 24 formulas	Design, demonstration, availability, and spare parts from Chapters 11, 12, and 13.	14 numbered · 10 named	**p. 22**

READING A CARD

`CRE-6.8` Numbered ID: equation 6.8 in the ASQ CRE Handbook.

`CRE-WEI` Named ID: a formula the handbook shows without an equation number.

`Handbook Ch. 6` The handbook chapter where the formula appears.

EACH CARD GIVES YOU

Equation Shaded panel, in the section colour.

When to use The situation the formula answers.

Exam trap The mistake that costs marks.

Formula index

Every formula by ID. Click any row to jump straight to that formula; every page footer links back here.

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CRE-FRAT F ratio of two chi-square variables 13
CRE-WELCH Two-sample t, unequal variances 13
CRE-POOL Two-sample t, pooled variance 13
CRE-ZP One-proportion z statistic 14
CRE-FPC Finite population correction 14
CRE-R3 Rule of 3 14
CRE-RLB Lower confidence bound on reliability 14
CRE-PCL Exact interval for a proportion 14
CRE-PCI Normal interval for a proportion 14
CRE-NPROP Sample size for a proportion 15
CRE-EXPM Confidence limits for an exponential mean 15
CRE-CP Process capability and performance, Cp and Pp 15
CRE-CPK Process capability and performance, Cpk and P... 15
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CRE-SSLN Lognormal stress-strength reliability 24
CRE-SN Taguchi signal-to-noise ratios 24
CRE-ANO ANOVA error sum of squares 24
CRE-LAM Sum of constant element failure rates 24
CRE-ALLOC Allocated repair time 25
CRE-SPR Poisson demand for spare parts 25
CRE-CRO Croston forecast for intermittent spares 25
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CRE BODY OF KNOWLEDGE · SECTION I

I Reliability Fundamentals

Life, failure, repair, and availability from Chapters 1 and 2, including the OEE and Duane displays.

19 FORMULAS

17 numbered · 2 named

CRE-2.1 Reliability at time t

LIFE AND FAILURE MEASURES · Handbook Ch. 2

$$R(t) = \Pr(X > t) = 1 - \Pr(X \leq t)$$

WHEN TO USE
Reliability at time t is the probability that life exceeds t.

EXAM TRAP
Do not report the probability of failure, Pr(X at or below t), as the reliability.

CRE-2.2 Empirical reliability at time t

LIFE AND FAILURE MEASURES · Handbook Ch. 2

$$R(t) = \frac{\text{units functioning at the end of the period}}{\text{units functioning at the start of the test}}$$

WHEN TO USE
Use the count still functioning divided by the count that started the test.

EXAM TRAP
Do not put the failed units in the numerator.

CRE-2.3 Mean time to failure

LIFE AND FAILURE MEASURES · Handbook Ch. 2

$$\text{MTTF} = \theta = \frac{\sum t_i}{n}$$

WHEN TO USE
For complete failure times, the mean is the total of the times divided by the number of items.

EXAM TRAP
Do not divide by the number of survivors.

CRE-2.4 Mean time between failures

LIFE AND FAILURE MEASURES · Handbook Ch. 2

$$\text{MTBF} = \frac{nm}{r}$$

WHEN TO USE
n items each exposed for m hours, with r failures, give total time nm divided by r.

EXAM TRAP
Do not divide by the number of items when the question asks for time per failure.

CRE-2.5 Mean cycles between failures

LIFE AND FAILURE MEASURES · Handbook Ch. 2

$$\text{MCBF} = \frac{nc}{r}$$

WHEN TO USE
The same total-exposure idea, counted in cycles instead of hours.

EXAM TRAP
Do not leave the answer in hours when the exposure is in cycles.

CRE-2.6 Mean time between critical failures

LIFE AND FAILURE MEASURES · Handbook Ch. 2

$$\text{MTBCF} = \frac{nc}{C}$$

WHEN TO USE
The denominator is the count of critical failures, not every failure.

EXAM TRAP
Using all failures in the denominator gives MTBF, not the critical-failure measure.

CRE-2.7 Constant failure rate

LIFE AND FAILURE MEASURES · Handbook Ch. 2

$$\lambda = \frac{1}{\text{MTBF}} = \frac{1}{\text{MTTF}}$$

WHEN TO USE
Under a constant rate, the rate is the reciprocal of the mean life.

EXAM TRAP
Do not multiply the mean by the rate and call that λ.

CRE-2.8 Mean time to repair

AVAILABILITY · Handbook Ch. 2

$$\text{MTTR} = \frac{\text{total repair time}}{\text{number of repair operations}}$$

WHEN TO USE
Average the observed repair times. This is a repair measure, not a life measure.

EXAM TRAP
Do not put operating hours in the numerator.

CRE-2.9 MTBF at a constant failure rate

LIFE AND FAILURE MEASURES | Handbook Ch. 2

$$\text{MTBF} = \frac{1}{\lambda}$$

WHEN TO USE
Invert a constant failure rate to get the mean time between failures.

EXAM TRAP
The rate itself is not a time.

CRE-2.10 MTTF at a constant failure rate

LIFE AND FAILURE MEASURES | Handbook Ch. 2

$$\text{MTTF} = \frac{1}{\lambda}$$

WHEN TO USE
The same reciprocal applies to a nonrepairable item's mean time to failure.

EXAM TRAP
Do not use this reciprocal when the hazard changes with age.

CRE-2.11 Mean downtime

AVAILABILITY | Handbook Ch. 2

$$\text{MDT} = \frac{\text{total downtime}}{\text{number of downtime occurrences}}$$

WHEN TO USE
Downtime includes more than active repair when the record says so.

EXAM TRAP
MTTR is the repair portion. Do not substitute it when the question asks for downtime.

CRE-2.12 Availability over an interval

AVAILABILITY | Handbook Ch. 2

$$A = \frac{\text{time the unit is capable of use}}{\text{length of the interval}}$$

WHEN TO USE
Compare capable time with the full interval named in the question.

EXAM TRAP
Do not omit scheduled or logistics time if the interval includes it.

CRE-2.13 Availability, general formula

AVAILABILITY | Handbook Ch. 2

$$A = \frac{\text{average uptime}}{\text{average uptime} + \text{average downtime}}$$

WHEN TO USE
Availability is uptime divided by uptime plus downtime.

EXAM TRAP
Leaving downtime out of the denominator produces 1.

CRE-2.14 Reliability at time t, constant failure rate

LIFE AND FAILURE MEASURES | Handbook Ch. 2

$$R(t) = e^{-\lambda t}$$

WHEN TO USE
A constant failure rate gives an exponential reliability.

EXAM TRAP
The exponent is negative. A positive exponent is greater than 1 and cannot be a reliability.

CRE-2.15 Availability from MTBF and MTTR

AVAILABILITY | Handbook Ch. 2

$$A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

WHEN TO USE
Use this when the only times given are the mean life and the mean repair.

EXAM TRAP
Do not put MTTR in the numerator.

CRE-2.16 Mean time between maintenance actions

AVAILABILITY | Handbook Ch. 2

$$\text{MTBMA} = \frac{1}{\lambda + \mu}$$

WHEN TO USE
λ is the failure rate and μ is the preventive maintenance rate. Both remove the item from service.

EXAM TRAP
Using only λ ignores the maintenance actions.

CRE-2.17 Availability from maintenance actions

AVAILABILITY · Handbook Ch. 2

$$A = \frac{\text{MTBMA}}{\text{MTBMA} + \text{MAMT}}$$

WHEN TO USE
MAMT is the mean active maintenance time that goes with MTBMA.

EXAM TRAP
Do not replace MAMT with MTTR unless the problem says they are the same.

CRE-OEE Overall equipment effectiveness

LIFE AND FAILURE MEASURES · Handbook Ch. 1

$$\text{OEE} = \text{Availability} \times \text{Performance} \times \text{Quality}$$

$$\text{Availability} = \frac{\text{actual operating time}}{\text{planned production time}}$$

$$\text{Performance} = \frac{\text{total pieces/actual operating time}}{\text{ideal pieces/time period}}$$

$$\text{Quality} = \frac{\text{good pieces}}{\text{total pieces}}$$

WHEN TO USE
OEE is the product of the three ratios. Planned production time is the shift less breaks. Actual operating time subtracts setup and maintenance downtime.

EXAM TRAP
Do not add the three ratios. A zero in any one of them makes OEE zero.

CRE-DUAN Duane cumulative-failure plot

LIFE AND FAILURE MEASURES · Handbook Ch. 1

$$\log_{10}\left(\frac{n(T)}{T}\right) = -\alpha \log_{10}(T) + b$$

WHEN TO USE
On log-log paper the cumulative failure rate n(T)/T is a straight line in cumulative time T.

EXAM TRAP
n(T) is the cumulative failure count, not the instantaneous rate. The slope is negative when reliability is improving.

CRE BODY OF KNOWLEDGE · SECTION III

III Probability & Statistics for Reliability

Probability, life distributions, sampling, capability, and control charts from Chapters 6 and 7.

56 FORMULAS

17 numbered · 39 named

CRE-6.1 Sample mean

DESCRIPTIVE STATISTICS AND REGRESSION · Handbook Ch. 6

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

WHEN TO USE
Add the observations and divide by how many there are.

EXAM TRAP
The divisor is n, not n minus 1. That correction belongs to the sample variance.

CRE-6.2 Pearson sample correlation

DESCRIPTIVE STATISTICS AND REGRESSION · Handbook Ch. 6

$$r = \frac{\mathrm{Cov}(X,Y)}{s_x s_y} = \frac{SS_{xy}}{\sqrt{SS_{xx}\,SS_{yy}}}$$

WHEN TO USE
Scale the sample covariance by the two sample standard deviations. The same ratio is the corrected sum of products over the square root of the two corrected sums of squares.

EXAM TRAP
A correlation is not a slope. The slope still has the units of Y per unit of X.

CRE-6.3 Simple linear regression model

DESCRIPTIVE STATISTICS AND REGRESSION · Handbook Ch. 6

$$y = \beta_0 + \beta_1 x + \varepsilon$$

WHEN TO USE
The model is a straight line plus a random error. The β values are unknown constants.

EXAM TRAP
Do not replace the error with the residual. The residual is what remains after fitting.

CRE-6.4 Fitted regression equation

DESCRIPTIVE STATISTICS AND REGRESSION · Handbook Ch. 6

$$\hat{y} = b_0 + b_1 x$$

WHEN TO USE
The fitted equation uses the estimated intercept and slope.

EXAM TRAP
A hat on y means a prediction, not an observed point.

CRE-6.5 Estimated intercept and slope

DESCRIPTIVE STATISTICS AND REGRESSION · Handbook Ch. 6

$$b_0 = \bar{y} - b_1 \bar{x}$$
$$b_1 = \frac{SS_{xy}}{SS_{xx}}$$

WHEN TO USE
Estimate the slope from the corrected sums of squares, then the intercept from the means.

EXAM TRAP
Swapping the sums of squares inverts the slope.

CRE-6.6 Estimated residuals

DESCRIPTIVE STATISTICS AND REGRESSION · Handbook Ch. 6

$$e_i = y_i - \hat{y}_i$$

WHEN TO USE
A residual is the observed response minus the fitted response.

EXAM TRAP
The reverse subtraction changes the sign of every residual.

CRE-6.7 Two-sided test statistic for the slope

DESCRIPTIVE STATISTICS AND REGRESSION · Handbook Ch. 6

$$t^* = \frac{b_1 - \beta_1}{s_{b_1}}$$

$$s_{b_1} = \frac{s_e}{\sqrt{SS_{xx}}}$$

$$s_e = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n-2}} = \sqrt{\frac{SSE}{n-2}}$$

WHEN TO USE

Compare the estimated slope with the hypothesized slope. The standard error uses the residual standard deviation and SS sub xx. Reject the null when the absolute value of t-star exceeds the two-sided critical value with n minus 2 degrees of freedom.

EXAM TRAP

The usual null value of beta sub 1 is 0, but only when the question says the slope is zero. The critical value uses alpha over 2 because the handbook states a two-sided rule.

CRE-6.8 Relative frequency

PROBABILITY · Handbook Ch. 6

$$\Pr(A) = \frac{a}{n}$$

WHEN TO USE

a is the number of times A occurred and n is the number of trials.

EXAM TRAP

This is an estimate of the probability, not a proof that the probability equals a over n.

CRE-6.9 Complement rule

PROBABILITY · Handbook Ch. 6

$$\Pr(\bar{A}) = 1 - \Pr(A)$$

WHEN TO USE

An event and its complement use up the whole probability.

EXAM TRAP

Do not subtract from the number of trials. Subtract from 1.

CRE-6.10 General addition rule

PROBABILITY · Handbook Ch. 6

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

WHEN TO USE

Add the two events, then subtract the overlap that was counted twice. The handbook's note after Equation 6.10 describes that overlap.

EXAM TRAP

The printed Equation 6.10 shows the union symbol on both sides. The sentence after it says the overlap was counted twice, so subtract the intersection, not the union.

CRE-6.11 Conditional probability

PROBABILITY · Handbook Ch. 6

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

WHEN TO USE

Restrict the denominator to the event that is known to have happened.

EXAM TRAP

Dividing by Pr(A) instead of Pr(B) answers a different conditional probability.

CRE-6.12 Probability of an intersection

PROBABILITY · Handbook Ch. 6

$$\Pr(A \cap B) = \Pr(B \mid A)\Pr(A) = \Pr(A \mid B)\Pr(B)$$

WHEN TO USE

The joint probability is either order: the probability of one event times the probability of the other given the first.

EXAM TRAP

Multiplying the two marginal probabilities is allowed only when the events are independent.

CRE-6.13 Permutation formula

PROBABILITY · Handbook Ch. 6

$$ {}_nP_r = \frac{n!}{(n-r)!} $$

WHEN TO USE
Use a permutation when the order of the r selected items matters.

EXAM TRAP
Dividing by r factorial as well is the combination, not the permutation.

CRE-6.14 Combination formula

PROBABILITY · Handbook Ch. 6

$$ {}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} $$

WHEN TO USE
Use a combination when the order of the r selected items does not matter.

EXAM TRAP
Stopping after the permutation counts the same group more than once.

CRE-7.1 Normalizing with a z transformation

DATA AND LIFE FUNCTIONS · Handbook Ch. 7

$$ \hat{z}_i = \frac{x_i - \bar{x}}{s} $$

WHEN TO USE
The handbook writes a hat on z. Subtract the sample mean and divide by the sample standard deviation.

EXAM TRAP
The handbook warns that this scaling does not make a non-normal sample normal.

CRE-7.2 Hazard over a time period

DATA AND LIFE FUNCTIONS · Handbook Ch. 7

$$ \lambda(t) = \frac{\text{fraction of failures during the period}}{\text{length of the period}} $$

WHEN TO USE
The period hazard is the failure fraction divided by the length of that period.

EXAM TRAP
Do not divide by the number of units that had already failed before the period.

CRE-7.3 Reliability function from survivors

DATA AND LIFE FUNCTIONS · Handbook Ch. 7

$$ R(t) = \frac{\text{number surviving at the end of the period}}{\text{number of units tested}} $$

WHEN TO USE
This is the empirical survivor fraction at the end of the period.

EXAM TRAP
The numerator is the survivors, not the failures.

CRE-VAR Sample variance and standard deviation

DESCRIPTIVE STATISTICS AND REGRESSION · Handbook Ch. 6

$$ s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2} $$

WHEN TO USE
The handbook's point estimate of variance divides by n minus 1.

EXAM TRAP
Equation 6.1 divides the mean by n. Using n here understates the variance.

CRE-BIN Binomial distribution

PROBABILITY · Handbook Ch. 6

$$ \Pr(X = x \mid n, p) = \binom{n}{x} p^x (1-p)^{n-x} $$

$$ \Pr(X \le x \mid n, p) = \sum_{k=0}^{x} \binom{n}{k} p^k (1-p)^{n-k} $$

WHEN TO USE
Use it for a fixed number of independent trials with the same success probability.

EXAM TRAP
The combination, not the permutation, is the coefficient. Order does not matter inside one count.

CRE-NBIN Negative binomial distribution

PROBABILITY · Handbook Ch. 6

$$ \Pr(X = n \mid r, p) = \binom{n-1}{r-1} p^r (1-p)^{n-r}, \quad n \ge r $$

$$ E(X) = \frac{r}{p} $$

WHEN TO USE
X is the trial number on which the r-th event occurs.

EXAM TRAP
The binomial counts events in a fixed n. This distribution lets the number of trials vary.

CRE-GEO Geometric distribution

PROBABILITY Handbook Ch. 6

$$\Pr(X = n) = p(1-p)^{n-1}, \qquad E(X) = \frac{1}{p}$$

WHEN TO USE
The geometric is the negative binomial with r equal to 1: trials until the first event.

EXAM TRAP
The exponent is n minus 1, not n. The mean is 1/p, not p.

CRE-HYP Hypergeometric distribution

PROBABILITY Handbook Ch. 6

$$\Pr(X = k \mid N, K, n) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

WHEN TO USE
A sample of n is drawn without replacement from N items, of which K are the type being counted.

EXAM TRAP
The binomial allows replacement. Without replacement, the probability changes after each draw.

CRE-POI Poisson distribution

PROBABILITY Handbook Ch. 6

$$\Pr(X = x \mid \lambda) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

$$\Pr(X \le x \mid \lambda) = \sum_{k=0}^{x} \frac{e^{-\lambda}\lambda^{k}}{k!}$$

WHEN TO USE
Use it for a count in a fixed window when the mean rate is constant.

EXAM TRAP
The mean is lambda, not the window length. Include lambda in both the power and the exponential.

CRE-EXP Exponential density and distribution

PROBABILITY Handbook Ch. 6

$$f(t) = \lambda e^{-\lambda t}$$

$$F(t) = 1 - e^{-\lambda t}$$

WHEN TO USE
The density is the constant rate times the reliability. The cdf is one minus that reliability.

EXAM TRAP
Equation 2.14 is the reliability e to the minus lambda t. Do not report the density as a reliability.

CRE-WEI Weibull distribution

PROBABILITY · Handbook Ch. 6

$$f(t) = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1}\exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right]$$

$$F(t) = 1 - \exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right]$$

$$R(t) = \exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right]$$

$$E(X) = \eta\Gamma\left(1+\frac{1}{\beta}\right)$$

$$\mathrm{SD}(X) = \eta\sqrt{\Gamma\left(1+\frac{2}{\beta}\right) - \left[\Gamma\left(1+\frac{1}{\beta}\right)\right]^2}$$

WHEN TO USE

Beta is the shape and eta is the scale. The mean multiplies the scale by a gamma value. The reference table also gives the standard deviation. It writes the gamma arguments as 2 over beta plus 1 and 1 over beta plus 1, which are the same values.

EXAM TRAP

At the characteristic life t equals eta, reliability is e to the minus 1, about 0.368, not 0.50.

CRE-NOR Normal distribution

PROBABILITY · Handbook Ch. 6

$$f(t) = \frac{1}{\sigma\sqrt{2\pi}}\exp\left[-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2\right]$$

$$F(t) = \Phi\left(\frac{t-\mu}{\sigma}\right)$$

$$Z = \frac{X-\mu}{\sigma}$$

WHEN TO USE

Standardize before reading a normal table. The handbook's reference table writes the same density with the square completed in the exponent.

EXAM TRAP

One integral display puts x, rather than the upper limit, inside Phi. Use the limit you actually integrated to.

CRE-LOGN Lognormal distribution

PROBABILITY · Handbook Ch. 6

$$\ln X \sim \mathrm{Normal}(\mu, \sigma^2)$$

$$f(x) = \frac{1}{x\sqrt{2\pi\sigma^2}}\exp\left[-\frac{(\ln x - \mu)^2}{2\sigma^2}\right]$$

$$E(X) = \exp\left(\mu + \frac{\sigma^2}{2}\right)$$

$$\mathrm{SD}(X) = \sqrt{\exp(2\mu+\sigma^2)\left(\exp(\sigma^2)-1\right)}$$

WHEN TO USE

Mu and sigma are the mean and standard deviation of the natural log, not of X. This is the reference-table density.

EXAM TRAP

A nearby display puts sigma squared in the denominator outside the square root. Use the reference-table form, with sigma inside the square root and x in front.

CRE-HAZ Hazard function

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$h(t) = \frac{f(t)}{R(t)}$$

WHEN TO USE

The hazard is the density among the units still surviving.

EXAM TRAP

Do not divide the density by the cdf. The denominator is the reliability.

CRE-HCU Cumulative hazard

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$H(t) = \int_0^t h(x)\,dx = -\ln R(t)$$

WHEN TO USE
The cumulative hazard is the area under the hazard, and it is also minus the log of reliability.

EXAM TRAP
Reliability is e to the minus H(t). Dropping the minus makes the result greater than 1.

CRE-COND Conditional reliability

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$R(t+M \mid t) = \frac{R(t+M)}{R(t)}$$

WHEN TO USE
Given survival to t, the probability of surviving an additional mission of length M is the ratio of the two reliabilities.

EXAM TRAP
Do not multiply R(t) by R(M) unless the hazard is constant. That product is the exponential special case.

CRE-KM Kaplan-Meier reliability

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$\hat{R}(t) = \prod_{i:\, t_i \le t} \left(1 - \frac{r_i}{n_i}\right)$$

WHEN TO USE
At each failure time, multiply by one minus the failures in that step divided by the number still at risk.

EXAM TRAP
Censored units leave the risk set. They are not counted as failures in r sub i.

CRE-WPP Weibull probability-plot line

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$\ln\ln\left(\frac{1}{1-F(t)}\right) = \beta \ln t - \beta \ln \eta$$

WHEN TO USE
A Weibull sample is a straight line when the vertical axis is the double log of one over one minus F.

EXAM TRAP
The slope is beta. Swapping the axes changes the slope to 1/beta.

CRE-CHI Chi-square relationship and Pearson statistic

DESCRIPTIVE STATISTICS AND REGRESSION · Handbook Ch. 6

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

$$X^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

WHEN TO USE
The first line is the sampling distribution of the variance. The second compares observed and expected counts.

EXAM TRAP
Expected counts belong in the denominator. Squared differences divided by observed counts are a different statistic.

CRE-T1 One-sample t statistic

DESCRIPTIVE STATISTICS AND REGRESSION · Handbook Ch. 6

$$t^* = \frac{\hat{X} - \mu_0}{\hat{\sigma}/\sqrt{n}} \sim t_{n-1}$$

WHEN TO USE
Compare the sample mean with the hypothesized mean, in standard-error units. A paired test is this same statistic on the differences.

EXAM TRAP
The divisor is s over square root of n, not s itself. The degrees of freedom are n minus 1.

CRE-MCI Confidence interval for a mean

DESCRIPTIVE STATISTICS AND REGRESSION · Handbook Ch. 6

$$\bar{X} \pm t_{\alpha/2,\, n-1}\sqrt{\frac{\hat{\sigma}^2}{n}}$$

$$\text{LCL} = \bar{X} - t_{\alpha,\, n-1}\sqrt{\frac{\hat{\sigma}^2}{n}}$$

WHEN TO USE

If sigma is known, the handbook sets sigma-hat equal to sigma and replaces t with z. Otherwise sigma-hat is the sample standard deviation s. A one-sided bound uses t with alpha, not alpha over 2, and n minus 1 degrees of freedom.

EXAM TRAP

The standard error is the square root of sigma-hat squared over n. Leaving the square root off divides by n and makes the interval too narrow.

CRE-FRAT F ratio of two chi-square variables

DESCRIPTIVE STATISTICS AND REGRESSION · Handbook Ch. 6

$$F^* = \frac{\chi_1^2/\nu_1}{\chi_2^2/\nu_2} \sim F_{\nu_1,\nu_2}$$

WHEN TO USE

An F for two variances is this ratio. Each chi-square is scaled by its own degrees of freedom.

EXAM TRAP

The larger sample variance is not automatically the numerator. The handbook puts the hypothesized direction in the numerator.

CRE-WELCH Two-sample t, unequal variances

DESCRIPTIVE STATISTICS AND REGRESSION · Handbook Ch. 6

$$t^* = \frac{(\hat{X}_1 - \hat{X}_2) - \mu_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

$$\nu = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{(s_1^2/n_1)^2/(n_1-1) + (s_2^2/n_2)^2/(n_2-1)}$$

WHEN TO USE

The handbook uses this Welch form, and the Satterthwaite degrees of freedom, when the two sigmas are not pooled.

EXAM TRAP

Do not use n1 plus n2 minus 2 with this standard error. Those degrees of freedom belong to the pooled test.

CRE-POOL Two-sample t, pooled variance

DESCRIPTIVE STATISTICS AND REGRESSION · Handbook Ch. 6

$$s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}}$$

$$t^* = \frac{(\hat{X}_1 - \hat{X}_2) - \mu_0}{s_p\sqrt{1/n_1 + 1/n_2}}$$

WHEN TO USE

Pool when the handbook says the variances may be combined. The degrees of freedom are n1 plus n2 minus 2.

EXAM TRAP

The pooled standard deviation is not the average of the two standard deviations.

CRE-ZP One-proportion z statistic

PROBABILITY · Handbook Ch. 6

$$z^* = \frac{\hat{p} - p_0}{\sqrt{\hat{p}(1-\hat{p})/n}}$$

$$\hat{p}_{AC} = \frac{x+1}{n+2}$$

WHEN TO USE

The handbook's displayed proportion statistic puts the estimated proportion in the standard error. The same table's Agresti-Caffo estimate adds one success and two trials before the proportion is formed.

EXAM TRAP

Some exams put the null proportion p0 under the square root instead. Read which standard error the question is using. The Agresti-Caffo proportion is not the z statistic's numerator.

CRE-FPC Finite population correction

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$\text{fpc} = \sqrt{\frac{N-n}{N-1}}$$

WHEN TO USE

Multiply a standard error by this factor when the sample is a noticeable part of a finite population.

EXAM TRAP

The correction is less than 1. Using it when the population is effectively infinite shrinks the interval without a reason.

CRE-R3 Rule of 3

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$n \approx \frac{3}{p}, \qquad (0,\ 3/n) \text{ for a small proportion}$$

WHEN TO USE

With no events expected, the sample size is about 3 divided by the proportion of interest. The matching approximate interval for a small p is 0 to 3/n.

EXAM TRAP

This is an approximation for a rare binomial event. Equation 11.1 is the exact zero-failure sample size.

CRE-RLB Lower confidence bound on reliability

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$R_L = \alpha^{1/n}$$

$$R_L = \frac{n-r}{(n-r) + (r+1)F_{\alpha,\nu_1,\nu_2}}$$

$$\nu_1 = 2(r+1), \qquad \nu_2 = 2(n-r)$$

WHEN TO USE

The first bound is the no-failure test. The second allows r failures and uses an F critical value.

EXAM TRAP

Alpha is the error rate, not the confidence. A 95 percent bound uses alpha equal to 0.05.

CRE-PCL Exact interval for a proportion

PROBABILITY · Handbook Ch. 6

$$\frac{\nu_1 F_{\alpha,\nu_1,\nu_2}}{\nu_2 + \nu_1 F_{\alpha,\nu_1,\nu_2}}$$

WHEN TO USE

The handbook's one-sided Clopper-Pearson limit has this form. For the upper limit, nu1 is 2(x+1) and nu2 is 2(n-x). For the lower limit, nu1 is 2x and nu2 is 2(n-x+1). A two-sided interval uses alpha over 2.

EXAM TRAP

The upper and lower limits do not use the same degrees of freedom.

CRE-PCI Normal interval for a proportion

PROBABILITY · Handbook Ch. 6

$$\hat{p} = \frac{x}{n}$$

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

WHEN TO USE

The handbook uses this Wald interval when x is at least 10 and n minus x is at least 10. A one-sided bound uses z with alpha instead of alpha over 2.

EXAM TRAP

This is not the exact Clopper-Pearson interval. That limit uses an F value and different degrees of freedom on each side. The z statistic for a hypothesis test is a different card.

CRE-NPROP Sample size for a proportion

DESIGN AND DEMONSTRATION · Handbook Ch. 6

$$n = \left(\frac{z_{\alpha/2}}{E}\right)^2 \hat{p}(1-\hat{p}) = \frac{1}{4}\left(\frac{z_{\alpha/2}}{E}\right)^2$$

WHEN TO USE

E is the margin of error. If the proportion is unknown, the handbook uses p equal to 0.5 because p times one minus p is then 1/4, the largest value. A one-sided margin uses z with alpha. Equation 11.2 is the matching sample size for a mean.

EXAM TRAP

Forgetting the square, or forgetting to multiply by p times one minus p, does not give this sample size. Do not reuse the mean formula, which has sigma in the numerator.

CRE-EXPM Confidence limits for an exponential mean

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$\hat{\theta} = \frac{T}{r}$$

$$\theta_{L,\,\text{time}} = \frac{2T}{\chi^2_{\alpha/2,\,\nu=2r+2}}$$

$$\theta_{L,\,\text{failure}} = \frac{2T}{\chi^2_{\alpha/2,\,\nu=2r}}$$

$$\theta_U = \frac{2T}{\chi^2_{1-\alpha/2,\,\nu=2r}}$$

WHEN TO USE

T is the total time on failed and unfailed units. A time-censored test ends at a chosen time and its lower limit uses 2r+2 degrees of freedom. A failure-censored test ends on a chosen failure and its lower limit uses 2r. The upper limit uses 2r for both. A one-sided limit uses alpha instead of alpha over 2.

EXAM TRAP

Swapping the chi-square tails swaps the lower and upper limits.

CRE-CP Process capability and performance, Cp and Pp

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$\hat{C}_p \text{ or } \hat{P}_p = \frac{\text{USL} - \text{LSL}}{6\hat{\sigma}}$$

$$\hat{\sigma} = \frac{\overline{mR}}{1.128} \quad (X\text{-}mR)$$

$$\hat{\sigma} = \frac{\bar{R}}{d_2} \quad (\bar{X}\text{-}R)$$

$$\hat{\sigma} = \frac{\bar{s}}{c_4} \quad (\bar{X}\text{-}s)$$

WHEN TO USE

The same width ratio is Cp with a short-term sigma and Pp with the overall standard deviation. The handbook estimates the short-term sigma from the chart in use: moving range over 1.128, R-bar over d2, or s-bar over c4. Performance uses the ordinary standard deviation of the whole data set.

EXAM TRAP

One worked display writes UCL minus LCL in this ratio. Those letters are control-chart limits. The specification width is USL minus LSL. Do not put the short-term sigma into Pp.

CRE-CPK Process capability and performance, Cpk and Ppk

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$\hat{C}_{pL} \text{ or } \hat{P}_{pL} = \frac{\bar{X} - \text{LSL}}{3\hat{\sigma}}$$

$$\hat{C}_{pU} \text{ or } \hat{P}_{pU} = \frac{\text{USL} - \bar{X}}{3\hat{\sigma}}$$

$$\hat{C}_{pk} \text{ or } \hat{P}_{pk} = \min(\hat{C}_{pL}, \hat{C}_{pU})$$

WHEN TO USE

Cpk and Ppk are the smaller of the two one-sided ratios. Capability uses a short-term sigma. Performance uses the overall standard deviation. The handbook writes the same minimum for the Ppk pair.

EXAM TRAP

Cp ignores where the mean sits. Cpk is smaller than Cp when the mean is off center. Do not report the short-term Cpk as Ppk.

CRE-TOL Statistical tolerance interval

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$\bar{x} \pm Ks$$

WHEN TO USE
K is the handbook's normal tolerance factor for the stated confidence and population coverage. It is not a t critical value.

EXAM TRAP
A confidence interval estimates the mean. A tolerance interval is built to cover a proportion of the individual units.

CRE-XBAR X-bar and R chart limits

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$\begin{aligned}
\mathrm{LCL}_{\bar{X}} &= \bar{\bar{X}} - A_2\bar{R} \\
\mathrm{UCL}_{\bar{X}} &= \bar{\bar{X}} + A_2\bar{R} \\
\mathrm{LCL}_{R} &= D_3\bar{R} \\
\mathrm{UCL}_{R} &= D_4\bar{R}
\end{aligned}$$

WHEN TO USE
Use the X-bar and R pair when the subgroup size is from 2 to 10.

EXAM TRAP
A2, D3, and D4 depend on the subgroup size. Do not reuse the constants from an individuals chart.

CRE-IMR Individuals and moving-range limits

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$\begin{aligned}
\mathrm{LCL}_{X} &= \bar{X} - E_2\overline{mR} \\
\mathrm{UCL}_{X} &= \bar{X} + E_2\overline{mR} \\
\mathrm{LCL}_{mR} &= D_3\overline{mR} \\
\mathrm{UCL}_{mR} &= D_4\overline{mR}
\end{aligned}$$

WHEN TO USE
Use the individuals chart when each subgroup has one observation.

EXAM TRAP
The center of the individuals chart is a single bar, not the double bar used on an X-bar chart.

CRE-XBARS X-bar and s chart limits

DATA AND LIFE FUNCTIONS · Handbook Ch. 6

$$\begin{aligned}
\mathrm{LCL}_{\bar{X}} &= \bar{\bar{X}} - A_3\bar{s} \\
\mathrm{UCL}_{\bar{X}} &= \bar{\bar{X}} + A_3\bar{s} \\
\mathrm{LCL}_{s} &= B_3\bar{s} \\
\mathrm{UCL}_{s} &= B_4\bar{s}
\end{aligned}$$

WHEN TO USE
Use X-bar and s when the subgroup size is greater than 10.

EXAM TRAP
One table cell labels the upper X-bar limit as a limit for R-bar. The worked example, and the matching lower limit, put A3 on the X-bar chart.

CRE-PCH p chart limits

PROBABILITY · Handbook Ch. 6

$$p_i = \frac{D_i}{n_i}$$

$$\mathrm{LCL}_{p} = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}}$$

$$\mathrm{UCL}_{p} = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}}$$

WHEN TO USE
The p chart plots the proportion defective. The limits change when the sample size changes.

EXAM TRAP
A negative calculated lower limit is plotted at zero. Do not extend it below zero.

CRE-NPCH np chart limits

PROBABILITY · Handbook Ch. 6

$$\mathrm{LCL}_{np} = n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})}$$

$$\mathrm{UCL}_{np} = n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})}$$

WHEN TO USE
The np chart plots the count defective when the sample size is constant.

EXAM TRAP
Do not use the np limits on a proportion. Divide by n first if the chart is a p chart.

CRE-CCH c chart limits

PROBABILITY Handbook Ch. 6

$$\mathrm{LCL}_c = \bar{c} - 3\sqrt{\bar{c}}, \qquad \mathrm{UCL}_c = \bar{c} + 3\sqrt{\bar{c}}$$

WHEN TO USE

The c chart counts defects on an inspection unit of constant size.

EXAM TRAP

If the opportunity size changes, the matching chart is the u chart, not the c chart.

CRE-UCH u chart limits

PROBABILITY Handbook Ch. 6

$$\mathrm{LCL}_u = \bar{u} - 3\sqrt{\frac{\bar{u}}{n_i}}$$

$$\mathrm{UCL}_u = \bar{u} + 3\sqrt{\frac{\bar{u}}{n_i}}$$

WHEN TO USE

The u chart plots defects per unit when the area of opportunity can change.

EXAM TRAP

One table cell prints the upper-limit subscript as w. The lower limit and the formula are the u chart.

IV

CRE BODY OF KNOWLEDGE · SECTION IV

Planning, Testing & Modeling

Growth, accelerated tests, and reliability models from Chapters 8, 9, and 10.

26 FORMULAS

15 numbered · 11 named

CRE-9.1 Arrhenius acceleration factor

ACCELERATED TESTING · Handbook Ch. 9

$$A_F = \exp\left[\left(\frac{E_A}{K}\right)\left(\frac{1}{T_U} - \frac{1}{T_S}\right)\right]$$

WHEN TO USE
Temperatures are absolute. Use temperature is the cooler use condition and stress temperature is the hotter test.

EXAM TRAP
Reversing the two temperatures changes the sign and makes the factor smaller than 1.

CRE-9.2 Power-law life model

ACCELERATED TESTING · Handbook Ch. 9

$$\frac{\text{life at rated stress}}{\text{life at accelerated stress}} = \left(\frac{\text{accelerated stress}}{\text{rated stress}}\right)^{b}$$

WHEN TO USE
Life shortens as stress rises when the exponent b is positive.

EXAM TRAP
Do not invert only one of the two ratios.

CRE-9.3 Arrhenius-Peck model

ACCELERATED TESTING · Handbook Ch. 9

$$A_F = \left(\frac{\text{humidity}_{\text{low}}}{\text{humidity}_{\text{high}}}\right)^{-2.66} \times \exp\left[\left(\frac{E_a}{K}\right)\left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right]$$

WHEN TO USE
Humidity and temperature both accelerate. The handbook writes the humidity power as minus 2.66.

EXAM TRAP
Dropping the humidity ratio leaves an ordinary Arrhenius factor.

CRE-9.4 Seeded-defect ratio

ACCELERATED TESTING · Handbook Ch. 9

$$\frac{\text{found unknown defects}}{\text{found known defects}} = \frac{\text{unfound unknown defects}}{\text{unfound known defects}}$$

WHEN TO USE
Known seeded defects estimate how many unknown defects remain.

EXAM TRAP
Do not treat the number found as the number that were there.

CRE-10.1 Series-system reliability

SYSTEM AND LIFE MODELS · Handbook Ch. 10

$$R_{\text{system}} = \prod_{i=1}^{n} R_i$$

WHEN TO USE
Every element must work. Multiply the independent reliabilities.

EXAM TRAP
Adding the reliabilities is not a series system.

CRE-10.2 Parallel-system reliability

SYSTEM AND LIFE MODELS · Handbook Ch. 10

$$R_{\text{system}} = 1 - \prod_{i=1}^{n}(1 - R_i)$$

WHEN TO USE
The system fails only when every element fails.

EXAM TRAP
The product of the reliabilities is the series answer, not the parallel answer.

CRE-10.3 K-out-of-N reliability

SYSTEM AND LIFE MODELS · Handbook Ch. 10

$$R_{\text{system}} = \sum_{i=k}^{n} \binom{n}{i} R^i (1-R)^{n-i}$$

WHEN TO USE
At least k of n identical independent elements must work.

EXAM TRAP
Stopping at exactly k misses the cases with more than k survivors.

CRE-10.4 Arrhenius life model

SYSTEM AND LIFE MODELS · Handbook Ch. 10

$$P(T \le t) = \Phi\left(t; \mu = \beta_0 + \frac{E_A}{KT}, \sigma\right)$$

WHEN TO USE
The location depends on absolute temperature through the Arrhenius term.

EXAM TRAP
Temperature in Celsius does not belong in the denominator.

CRE-10.5 Arrhenius model, alternate form

SYSTEM AND LIFE MODELS | Handbook Ch. 10

$$t_p(T) = A_0 \exp\left[\frac{E_A}{KT}\right]$$

WHEN TO USE
This writes a percentile life directly as a function of absolute temperature.

EXAM TRAP
The exponential is greater at a lower temperature, so a hotter test has a shorter life.

CRE-10.6 Arrhenius acceleration factor from percentile life

SYSTEM AND LIFE MODELS | Handbook Ch. 10

$$AF(T) = \frac{t_p(T_U)}{t_p(T_S)} = \exp\left[\frac{E_A}{K}\left(\frac{1}{T_U} - \frac{1}{T_S}\right)\right]$$

WHEN TO USE
The factor is use life divided by stress life.

EXAM TRAP
Dividing in the other order estimates how much the test is slower, not faster.

CRE-10.7 Inverse-power acceleration factor

SYSTEM AND LIFE MODELS | Handbook Ch. 10

$$AF(V) = \frac{V^b}{V_U^b}$$

WHEN TO USE
The handbook's alternate voltage form is the stress voltage to the power b, divided by the use voltage to that power.

EXAM TRAP
Forgetting the exponent treats life as proportional to voltage.

CRE-10.8 Eyring model, temperature and one stress

SYSTEM AND LIFE MODELS | Handbook Ch. 10

$$t_f = AT^{\alpha} \exp\left[\frac{\Delta H}{kT} + \left(B + \frac{C}{T}\right)S_1\right]$$

WHEN TO USE
Temperature appears both as a power and inside the exponential, and S1 is the added stress.

EXAM TRAP
Setting the stress term to zero does not remove the temperature power.

CRE-10.9 Expanded Eyring model

SYSTEM AND LIFE MODELS | Handbook Ch. 10

$$t_f = AT^{\alpha} \exp\left[\frac{\Delta H}{kT} + \left(B + \frac{C}{T}\right)S_1 + \left(D + \frac{E}{T}\right)S_2\right]$$

WHEN TO USE
A second stress adds one more term of the same form.

EXAM TRAP
Do not add the two stresses before multiplying by their own coefficients.

CRE-10.10 Eyring factor relative to Arrhenius

SYSTEM AND LIFE MODELS | Handbook Ch. 10

$$AF_{\text{Ey}} = \left(\frac{\text{Temp}_K}{\text{Temp}_{U,K}}\right)^m AF_{\text{Ar}}$$

WHEN TO USE
Both temperatures are in kelvin. The Eyring factor is the Arrhenius factor times a power of that temperature ratio.

EXAM TRAP
The power is not a second activation energy. Over ordinary temperature ranges the handbook often drops it and keeps Arrhenius.

CRE-10.11 Reliability, proportional hazards model

SYSTEM AND LIFE MODELS | Handbook Ch. 10

$$R(t) = e^{-\lambda(x)\,t}$$

WHEN TO USE
After the stresses are folded into the failure rate $\lambda(x)$, reliability at time t is $e^{-\lambda(x)t}$.

EXAM TRAP
Do not leave the stresses out of the rate. $\lambda(x)$ already carries them, and this is not a series-product formula.

CRE-GROW Reliability growth slope and instantaneous MTBF

SYSTEM AND LIFE MODELS | Handbook Ch. 8

$$b = \frac{\log(\theta_1/\theta_0)}{\log(T_1/T_0)}, \qquad \theta = \frac{\theta_m}{1-b}$$

WHEN TO USE
b is the growth slope between two cumulative times. Theta sub m is the cumulative MTBF, and theta is the instantaneous MTBF.

EXAM TRAP
The instantaneous value is larger than the cumulative MTBF while reliability is still growing. Do not report theta sub m as the current rate.

CRE-W0 Weibull zero-failure sample size

DESIGN AND DEMONSTRATION · Handbook Ch. 8

$$n = \frac{1}{k^{\beta}} \frac{\log \alpha}{\log R}$$

WHEN TO USE
k is the number of required lives the test runs, beta is the Weibull shape, alpha is one minus the confidence, and R is the reliability to demonstrate.

EXAM TRAP
Equation 11.1 is the exponential case. Leaving out 1 over k to the beta ignores a test that runs longer than one life.

CRE-ARATE Arrhenius reaction rate

ACCELERATED TESTING · Handbook Ch. 9

$$R = A \exp\left(-\frac{E_A}{KT}\right)$$

WHEN TO USE
This R is the reaction rate at absolute temperature T. The acceleration factor is the ratio of two such rates.

EXAM TRAP
Do not read this R as reliability. Equation 9.1 is the acceleration factor, and a reliability cannot contain the positive constant A in front.

CRE-AFDEF Acceleration factor as a rate ratio

ACCELERATED TESTING · Handbook Ch. 9

$$A_F = \frac{R_S}{R_U}$$

WHEN TO USE
The factor is the rate at the stress condition divided by the rate at the use condition. Equation 9.1 writes that ratio for the Arrhenius rate.

EXAM TRAP
Inverting the ratio produces a factor below 1 when the stress test is harsher.

CRE-IPL Inverse-power life

SYSTEM AND LIFE MODELS · Handbook Ch. 10

$$L = \frac{a}{V^b}, \qquad a > 0,\ b > 0$$

WHEN TO USE
Life falls as voltage, or another stress V, rises. Equation 10.7 is the acceleration factor that follows from this life model.

EXAM TRAP
The stress belongs in the denominator. Putting it in the numerator makes life increase with stress.

CRE-PHAZ Proportional-hazards failure rate

SYSTEM AND LIFE MODELS · Handbook Ch. 10

$$\lambda(x) = \sum_{i=0}^{k} a_i x_i$$

WHEN TO USE
The handbook's simplest proportional-hazards rate is a linear sum of the stresses. Equation 10.11 then gives reliability e to the minus lambda of x times t.

EXAM TRAP
Do not drop a stress that the model includes, and do not treat this sum itself as a reliability.

CRE-BEAR Ball-bearing multiplicative failure rate

SYSTEM AND LIFE MODELS · Handbook Ch. 10

$$\lambda = \lambda_b \left(\frac{L_a}{L_S}\right)^{y} \left(\frac{A_C}{0.006}\right)^{2.36} \left(\frac{v_0}{v_1}\right)^{0.54} \left(\frac{C_1}{60}\right)^{0.67} \left(\frac{M_b}{M_f}\right) C_w$$

WHEN TO USE
The handbook prints this product as a ball-bearing failure-rate model. The powers 2.36, 0.54, and 0.67 are part of that printed model.

EXAM TRAP
Those powers are not universal acceleration exponents. Do not reuse them for a voltage or temperature model.

CRE-FAT Fatigue life distribution

SYSTEM AND LIFE MODELS · Handbook Ch. 10

$$P(N \le n) = \Phi\left(n;\ \mu = \frac{\beta_0}{B} - \frac{1}{B}\log s,\ \sigma = \frac{\sigma_X}{B}\right)$$

WHEN TO USE
This is the handbook's normal distribution for fatigue life N at stress s.

EXAM TRAP
Equation 10.11 is the proportional-hazards reliability. It is not this fatigue formula.

CRE-WINV Weibull inverse for a simulated life

SYSTEM AND LIFE MODELS · Handbook Ch. 10

$$X_i = -\eta[\ln(1 - P_i)]^{1/\beta}$$

WHEN TO USE
P sub i is a random probability. The handbook uses this inverse to turn that probability into a Weibull life.

EXAM TRAP
The sign in front of eta is negative because the log of a probability is negative. Dropping the sign produces a negative life.

CRE-DSTEP System failure by conditioning

SYSTEM AND LIFE MODELS | Handbook Ch. 10

$$F_S = F_{S|C_1} R_{C_1} + F_{S|C_1^c} F_{C_1}$$

WHEN TO USE

Condition on whether the chosen component works, then weight the two system-failure probabilities by that component's reliability and unreliability.

EXAM TRAP

This is not the parallel formula. The parallel formula is Equation 10.2.

CRE-MARK Two-state repair transition matrix

SYSTEM AND LIFE MODELS | Handbook Ch. 10

$$T = \begin{bmatrix} 1 - P_{S-F} & P_{S-F} \\ P_{F-S} & 1 - P_{F-S} \end{bmatrix}$$

WHEN TO USE

Rows are the state now and columns are the state next. S-F is the chance of going from success to failure.

EXAM TRAP

Each row sums to 1. Replacing a diagonal entry with the off-diagonal probability double-counts that row.

V

CRE BODY OF KNOWLEDGE · SECTION V

Lifecycle Reliability

Design, demonstration, availability, and spare parts from Chapters 11, 12, and 13.

24 FORMULAS

14 numbered · 10 named

CRE-11.1 Sample size for a zero-failure test

DESIGN AND DEMONSTRATION · Handbook Ch. 11

$$n = \frac{\ln(1 - \text{confidence})}{\ln(\text{reliability})}$$

WHEN TO USE

Both the confidence complement and the reliability are below 1, so both logarithms are negative and the sample size is positive.

EXAM TRAP

Using the confidence itself, instead of one minus the confidence, changes the numerator. Chapter 8 writes the same exponential sample size as log of alpha over log of R, with alpha equal to one minus the confidence.

CRE-11.2 Sample size for a continuous distribution

DESIGN AND DEMONSTRATION · Handbook Ch. 11

$$n = \left(\frac{Z\sigma}{E}\right)^2$$

WHEN TO USE

Z is the normal factor, σ is the standard deviation, and E is the margin.

EXAM TRAP

The ratio is squared. Stopping before the square understates the sample.

CRE-11.3 Factor of safety

DESIGN AND DEMONSTRATION · Handbook Ch. 11

$$\text{factor of safety} = \frac{\text{strength}}{\text{stress}}$$

WHEN TO USE

Divide the strength by the stress applied.

EXAM TRAP

A factor below 1 means the stress is larger than the strength.

CRE-11.4 Margin of safety

DESIGN AND DEMONSTRATION · Handbook Ch. 11

$$\text{margin of safety} = \frac{\text{strength}}{\text{stress}} - 1$$

WHEN TO USE

The margin is the factor of safety minus 1.

EXAM TRAP

Reporting the factor of safety itself overstates the margin by 1.

CRE-12.1 Availability from MTBF and MTTR

MAINTAINABILITY AND SPARES · Handbook Ch. 12

$$A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

WHEN TO USE

The handbook's worked example divides MTBF by MTBF plus MTTR.

EXAM TRAP

The title mentions MTTF. Use the mean the problem actually gives, and keep the repair time in the denominator.

CRE-13.1 Inherent availability

MAINTAINABILITY AND SPARES · Handbook Ch. 13

$$A_i = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

WHEN TO USE

Inherent availability uses active repair time only.

EXAM TRAP

Logistics delay does not belong in this denominator.

CRE-13.2 Achieved availability

MAINTAINABILITY AND SPARES | Handbook Ch. 13

$$A_a = \frac{\text{MTBM}}{\text{MTBM} + \text{MMT}}$$

WHEN TO USE
Achieved availability uses the mean time between maintenance and the mean maintenance time.

EXAM TRAP
MTBF and MTTR answer the inherent question, not this one.

CRE-13.3 Operational availability

MAINTAINABILITY AND SPARES | Handbook Ch. 13

$$A_o = \frac{\text{uptime}}{\text{uptime} + \text{downtime}}$$

WHEN TO USE
Operational availability keeps every delay that is part of downtime.

EXAM TRAP
Removing logistics delay turns the result into an inherent or achieved measure.

CRE-13.4 MTBF from an operating record

MAINTAINABILITY AND SPARES | Handbook Ch. 13

$$\text{MTBF} = \frac{\text{operating hours}}{\text{failures}}$$

WHEN TO USE
Divide the operating hours by the number of failures in that record.

EXAM TRAP
Calendar hours are too large when the item was not operating the whole time.

CRE-13.5 MTBF from the failure rate

MAINTAINABILITY AND SPARES | Handbook Ch. 13

$$\text{MTBF} = \frac{1}{\lambda}$$

WHEN TO USE
Invert the constant failure rate.

EXAM TRAP
The failure rate is not already a mean time.

CRE-13.6 System mean repair time

MAINTAINABILITY AND SPARES | Handbook Ch. 13

$$\text{MTTR}_{\text{system}} = \frac{\sum_{i=1}^{n} \lambda_i t_i}{\sum_{i=1}^{n} \lambda_i}$$

WHEN TO USE
Weight each element's repair time by its failure rate.

EXAM TRAP
A plain average treats a rare failure as equal to a frequent one.

CRE-13.7 Moving average for spare parts

MAINTAINABILITY AND SPARES | Handbook Ch. 13

$$R_t = \frac{X_{t-1} + X_{t-2} + \cdots + X_{t-n}}{n}$$

WHEN TO USE
The forecast is the average of the last n demands.

EXAM TRAP
The current period is not in the average. The newest demand used is the previous period.

CRE-13.8 Weighted moving average for spare parts

MAINTAINABILITY AND SPARES | Handbook Ch. 13

$$R_t = \frac{nX_{t-1} + (n-1)X_{t-2} + \cdots + X_{t-n}}{n + (n-1) + \cdots + 1}$$

WHEN TO USE
More recent demands get the larger weights, and the weights in the denominator sum to the same total.

EXAM TRAP
Using n as the denominator ignores the smaller weights.

CRE-13.9 Single exponential smoothing for spare parts

MAINTAINABILITY AND SPARES | Handbook Ch. 13

$$R_t = \alpha X_{t-1} + (1-\alpha) R_{t-1}$$

WHEN TO USE
α is the smoothing weight, usually 0.1 to 0.4. A larger α follows recent demand more closely.

EXAM TRAP
α is not the failure rate.

CRE-SSN Normal stress-strength interference

DESIGN AND DEMONSTRATION · Handbook Ch. 11

$$\mu_D = \mu_Y - \mu_X$$
$$\sigma_D = \sqrt{\sigma_Y^2 + \sigma_X^2}$$
$$\Pr(\text{failure}) = \Pr\left(Z < \frac{0 - \mu_D}{\sigma_D}\right)$$

WHEN TO USE
Y is strength and X is stress. Failure is the event that their difference is negative. The handbook treats the two as independent normals.

EXAM TRAP
The standard deviation of the difference adds the variances, not the standard deviations.

CRE-SSEX Exponential stress-strength reliability

DESIGN AND DEMONSTRATION · Handbook Ch. 11

$$\Pr(D > 0) = \frac{\beta_Y}{\beta_X + \beta_Y}$$

WHEN TO USE
For independent exponential stress and strength, reliability is the strength mean divided by the sum of the two means.

EXAM TRAP
Failure probability is the stress mean over that same sum, and that equals one minus this reliability. Do not subtract the strength mean itself.

CRE-SSLN Lognormal stress-strength reliability

DESIGN AND DEMONSTRATION · Handbook Ch. 11

$$m_X = e^{\mu_X}, \qquad m_Y = e^{\mu_Y}$$
$$\Pr(D > 0) = \Pr\left(Z < \frac{\ln(m_Y / m_X)}{\sqrt{s_X^2 + s_Y^2}}\right)$$

WHEN TO USE
m is the median, e to the mu. s is the standard deviation on the log scale.

EXAM TRAP
Using the ordinary means of X and Y in the log ratio does not match this formula.

CRE-SN Taguchi signal-to-noise ratios

DESIGN AND DEMONSTRATION · Handbook Ch. 11

$$\text{larger is better: } S/N_i = -10 \log\left(\frac{1}{n_j}\sum_{j=1}^{n_j}\frac{1}{y_{ij}^2}\right)$$
$$\text{smaller is better: } S/N_i = -10 \log\left(\frac{1}{n_j}\sum_{j=1}^{n_j} y_{ij}^2\right)$$
$$\text{nominal is best: } S/N_i = 10 \log\left(\frac{\bar{y}_i^2}{s_i^2}\right)$$

WHEN TO USE
Pick the form from the direction of the response. The nominal form uses the square of the mean over the variance.

EXAM TRAP
The larger-is-better sum is of reciprocal squares. Using the raw squares is the smaller-is-better ratio.

CRE-ANO ANOVA error sum of squares

DESIGN AND DEMONSTRATION · Handbook Ch. 11

$$SS_E = SS_T - SS_{\text{Treatments}}$$
$$SS_E = SS_T - SS_A - SS_B - SS_{AB}$$

WHEN TO USE
The first line is the one-factor design. The second is the two-factor design with interaction. Degrees of freedom subtract in the same pattern.

EXAM TRAP
A block design also subtracts the block sum of squares. Leaving a term out inflates the error.

CRE-LAM Sum of constant element failure rates

MAINTAINABILITY AND SPARES · Handbook Ch. 13

$$\lambda = \sum_{i=1}^{n} \lambda_i$$

WHEN TO USE
When every element must work, the constant system rate is the sum of the element rates.

EXAM TRAP
Equation 10.1 multiplies reliabilities. Do not multiply the rates.

CRE-ALLOC Allocated repair time

MAINTAINABILITY AND SPARES | Handbook Ch. 13

$$t_i^* = \frac{t_i}{t}\, t^*$$

WHEN TO USE

The current repair time of element i is scaled by the ratio of the target system repair time t-star to the current system repair time t.

EXAM TRAP

Equation 13.6 builds the system repair time. This display allocates a new target back to the elements.

CRE-SPR Poisson demand for spare parts

MAINTAINABILITY AND SPARES | Handbook Ch. 13

$$\Pr(r) = \sum_{n=0}^{r} \frac{(\lambda t)^n e^{-\lambda t}}{n!}$$

WHEN TO USE

The mean number of demands in time t is lambda t. The sum is the probability of r or fewer demands.

EXAM TRAP

Lambda alone is not the Poisson mean unless the window length is 1.

CRE-CRO Croston forecast for intermittent spares

MAINTAINABILITY AND SPARES | Handbook Ch. 13

$$Z_{t+1} = \alpha X_t + (1-\alpha) Z_t$$
$$P_{t+1} = \beta G_t + (1-\beta) P_t$$
$$R_{t+1} = \frac{Z_{t+1}}{P_{t+1}}$$

WHEN TO USE

When demand is positive, smooth the size Z and the gap P, then divide those updated values. If the latest demand is zero, the handbook leaves Z, P, and the forecast unchanged. Example 13.8 uses alpha 0.2 and beta 0.3. After a demand of 16 with a gap of 2, the next size is 21.60 and the next gap is 1.30, and 21.60 divided by 1.30 is 16.62.

EXAM TRAP

The printed forecast line writes Z sub t over P sub t. The example header and the week-4 result divide the values just updated. Equation 13.9 smooths every period. Croston waits for a nonzero demand.

CRE-HUM Human performance reliability

MAINTAINABILITY AND SPARES | Handbook Ch. 12

$$R_e(t) = \exp\left[-\int_0^t h_e(t)\, dt\right]$$

WHEN TO USE

h sub e is the human error rate. Reliability follows the same identity as the cumulative hazard.

EXAM TRAP

The handbook display writes this exponential without the minus. Cumulative hazard earlier in the book is the integral of the hazard, and reliability is e to that integral with a minus. The printed plus sign would make reliability greater than 1.